

YANXIANG ZHU

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EDUCATION

Columbia University

New York, NY

M.S. in Electrical Engineering

Sep 2026 – Dec 2027

- Selected Coursework (*In Progress*): Digital VLSI Circuits, Formal Verification, Analog Electronics Circuits

Fudan University

Shanghai, China

B.S. in Microelectronics

Sep 2022 – Jun 2026

- Fudan First-Class Scholarship (*Top 3% of students*), NITORI International Scholarship, Fung Scholar
- GPA: **92/100**
- Selected Coursework: System on Chip, Digital Integrated Circuit, VLSI Technology, ASIC Design Methodology

University of Hong Kong

Hong Kong SAR, China

Exchange Student

Aug 2024 – Dec 2024

- Selected Coursework: Computer Architecture, Digital System, Computer Vision, Natural Language Processing

SKILLS

Programming Languages: (Hardware) Verilog, VHDL, Chisel; (Software) Python, C, C++, MATLAB;

Frameworks & Design Tools: FireSim, Petalinux, Vivado, ModelSim, Synopsys Design Compiler, Synopsys IC Compiler

RESEARCH

Hardware-Software Co-Acceleration for World Action Models

New York, NY

Research Assistant, Directed by Prof. Zishen Wan

Aug 2026 – now

- Built a modular WAM testbed supporting interchangeable model architectures for profiling and acceleration.

Full-Stack Simulation Framework for Intelligent Control Systems

Shanghai, China

Research Assistant, Directed by Prof. Jun Han

Jun 2025 – Apr 2026

- Extended UCB's RoSE framework to a customized SoC platform on a Xilinx Alveo V80 FPGA by redesigning host-FPGA communication and system software, enabling end-to-end closed-loop pre-silicon co-simulation with AirSim and system-level validation of custom accelerator IP for realistic UAV workloads.

Stochastic Computing Neural Network Accelerator

Shanghai, China

Leading Researcher, Directed by Prof. Jiang Xiao

Sep 2023 – Apr 2026

- Developed stochastic-sequence neural networks and bit-stream operators to reduce the arithmetic cost of conventional inference, replacing multiply-accumulate operations with lightweight bit-wise computation.
- Optimized stochastic encoding, nonlinear operators, and memory organization for FPGA deployment, achieving accuracy within 1% of conventional methods without floating-point arithmetic and with 10x lower hardware cost.

INTERNSHIP

Mammoth Supercomputing Semiconductor

Shanghai, China

Digital IC Design Intern — AI Accelerator Team

Apr 2026 – Jul 2026

- Designed the top-level architecture and RTL compute data path for a CNN super-resolution inference ASIC, integrating DDR4, on-chip SRAM, convolution acceleration, and host interfaces into a complete SoC.
- Designed a 128KB dual-port SRAM with ping-pong buffering to hide memory latency, and implemented hardware-controlled sliding-window dataflow for systolic-array convolution, sustaining full compute utilization.
- Validated end-to-end super-resolution workloads on FPGA before tape-out using the TWM compiler toolchain, resolving instruction, memory, and dataflow issues through cycle-level waveform debugging.

OPEN-SOURCE

Advanced RISC-V CPU Design on FPGA

Feb 2025 – Aug 2025

- Built a 5-stage pipelined RISC-V CPU with RV32I ISA support with exception handling, branch prediction (BHT+BTB), instruction forwarding, out-of-order issue, and novel memory architecture, validated by Dhrystone tests and real-time UART control using FPGA.